

Basic probability theory

C

This appendix reviews briefly the basic properties of random numbers and the probabilistic framework. This includes a list of some common *probability density functions* used in this book. This section is not about statistics, which is primarily concerned with hypothesis testing. Rather, we review here some concepts that provide a useful description framework of brain processes.

C.1 Random numbers and their probability (density) function

The most important concept introduced in this appendix is that of *random numbers*, which are denoted by capital letters in this appendix to distinguish them from regular numbers written in lower case. A random variable, X , is a quantity that can have different values each time the variable is inspected, such as in measurements in experiments. This is fundamentally different to a regular variable, x , which does not change its value once it is assigned. A random number is thus a new mathematical concept, not included in the regular mathematics of numbers. A specific value of a random number is still meaningful as it might influence specific processes in a deterministic way. However, since a random number can change every time it is inspected, it is also useful to describe more general properties when drawing examples many times. The frequency with which numbers can occur is then the most useful quantity to take into account. This frequency is captured by the mathematical construct of a *probability*.

We can formalize this with some compact notations. We speak of a *discrete random variable* in the case of discrete numbers for the possible values of a random number. A *continuous random variable* is a random variable that has possible values in a continuous set of numbers. There is, in principle, not much difference between these two kinds of random variables, except that the mathematical formulation has to be slightly different to be mathematically correct. For example, the *probability function*,

$$P_X(x) = P(X = x) \quad (\text{C.1})$$

describes the frequency with which each possible value x of a discrete variable X occurs. Note that x is a regular variable, not a random variable. The value of $P_X(x)$ gives the fraction of the times we get a value x for the random variable X if we draw many examples of the random variable.¹ From this definition it follows that the frequency of having any of the possible values is equal to one,

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¹Probabilities are sometimes written as a percentage, but we will stick to the fractional notation.

which is the normalization condition

$$\sum_x P_X(x) = 1. \quad (\text{C.2})$$

In the case of continuous random numbers we have an infinite number of possible values x so that the fraction for each number becomes infinitesimally small. It is then appropriate to write the probability distribution function as $P_X(x) = p_X(x)dx$, where $p_X(x)$ is the *probability density function* (pdf). The sum in eqn C.2 then becomes an integral, and normalization condition for a continuous random variable is

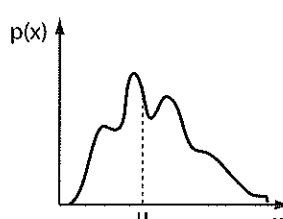
$$\int_x p_X(x)dx = 1. \quad (\text{C.3})$$

We will formulate the rest of this section in terms of continuous random variables. The corresponding formulas for discrete random variables can easily be deduced by replacing the integrals over the pdf with sums over the probability function. It is also possible to use the δ -function, outlined in Appendix A, to write discrete random processes in a continuous form.

C.2 Moments: mean, variance, etc.

In the following we only consider independent random values that are drawn from identical pdfs, often labeled as iid (independent and identically distributed) data. That is, we do not consider cases where there is a different probabilities of getting certain numbers when having a specific number in a previous trial. The static probability density function describes, then, all we can know about the corresponding random variable.

Let us consider the arbitrary pdf, $p_X(x)$, with the following graph:



Such a distribution is called *multimodal* because it has several peaks. Since this is a pdf, the area under this curve must be equal to one, as stated in eqn C.3. It would be useful to have this function parameterized in an analytical format. Most pdfs have to be approximated from experiments, and a common method is then to fit a function to the data. However, sometimes it is sufficient to know at least some basic characteristics of the functions. For example, we might ask what the most frequent value is when drawing many examples. This number is given by the largest peak value of the distribution. A more common quantity

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to know is the expected arithmetic average of those numbers, which is called the *mean*, *expected value*, or *expectation value* of the distribution, defined by

$$\mu = \int_{-\infty}^{\infty} xf(x)dx. \quad (C.4)$$

In the discrete case, this formula corresponds to the formula of calculating an arithmetic average, where we add up all the different numbers together with their frequency. Formally, we need to distinguish between a quantity calculated from random numbers and quantities calculated from the pdfs. If we treat the pdf as fundamental, then the arithmetic average is like an estimation of the mean. This is usually how it is viewed. However, we could also be pragmatic and say that we only have a collection of measurements so that the numbers are the 'real' thing, and that pdfs are only a mathematical construct. While this is mainly a philosophical debate, we try to be consistent in calling the quantities derived from data 'estimates' of the quantities defined through pdfs.

The mean of a distribution is not the only interesting quantity that characterizes a distribution. For example, we might want to ask what the *median* value is for which it is equally likely to find a value lower or larger than this value. Furthermore, the spread of the pdf around the mean is also very revealing as it gives us a sense of how spread the values are. This spread is often characterized by the standard deviation (std), or its square, which is called *variance*, σ^2 , and is defined as

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx. \quad (C.5)$$

This quantity is generally not enough to characterize the probability function uniquely; this is only possible if we know all moments of a distribution, where the *n*th moment about the mean is defined as

$$m^n = \int_{-\infty}^{\infty} (x - \mu)^n f(x)dx. \quad (C.6)$$

The *variance* is the second moment about the mean,

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx. \quad (C.7)$$

Higher moments specify further characteristics such as the kurtosis and skewness of the distribution. Moments higher than this have not been given explicit names. Knowing all moments of a distribution is equivalent in knowing the distribution precisely, and knowing a pdf is equivalent in knowing everything we could know about a random variable.

In case the distribution function is not given, moments have to be estimated from data. For example, the mean can be estimated from a sample of measurements by the *sample mean*,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad (C.8)$$

and the variance either from the *biased sample variance*,

$$s_1^2 = \frac{1}{n} \sum_{i=1}^n (\bar{x} - x_i)^2, \quad (C.9)$$

or the *unbiased sample variance*

$$s_2^2 = \frac{1}{n-1} \sum_{i=1}^n (\bar{x} - x_i)^2. \quad (\text{C.10})$$

A statistic is said to be biased if the mean of the sampling distribution is not equal to the parameter that is intended to be estimated. Knowing all moments uniquely specifies a pdf.

C.3 Examples of probability (density) functions

There is an infinite number of possible pdfs. However, some specific forms have been very useful for describing some specific processes and have thus been given names. The following is a list of some frequently used examples.

C.3.1 Bernoulli distribution

A Bernoulli random variable is a variable from an experiment that has two possible outcomes: success with probability p ; or failure, with probability $(1 - p)$.

Probability function:

$$P(\text{success}) = p; P(\text{failure}) = 1 - p$$

mean: p

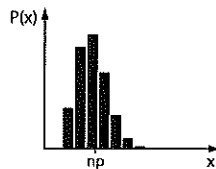
variance: $p(1 - p)$

C.3.2 Binomial distribution

The number of successes in n Bernoulli trials with probability of success p is binomially distributed. Note that the binomial coefficient is defined as

$$\binom{n}{x} = \frac{n!}{x!(n-x)!} \quad (\text{C.11})$$

and is given by the MATLAB function `nchoosek`.



Probability function:

$$P(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

mean: np

variance: $np(1-p)$

C.3.3 Chi-square distribution

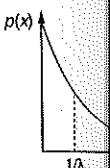
The sum of the squares of normally distributed random numbers is chi-square distributed and depends on a parameter ν that is equal to the mean. Γ is the

gamma func



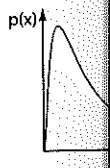
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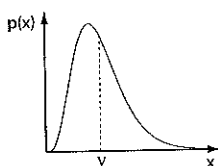
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gamma function included in MATLAB as `gamma`.

(C.10)



Probability density function:

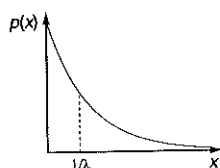
$$p(x) = \frac{x^{(\nu-2)/2} e^{-x/2}}{2^{\nu/2} \Gamma(\nu/2)}$$

mean: ν

variance: 2ν

C.3.4 Exponential distribution

Distributions of time between events that are Poisson distributed. Depends on one parameter λ , which is called the hazard function.



Probability density function:

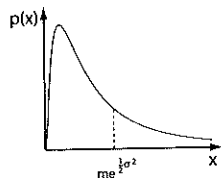
$$p(x) = \lambda e^{-\lambda x}$$

mean: $1/\lambda$

variance: $1/\lambda^2$

C.3.5 Lognormal distribution

Distribution that is constrained to be zero at $x = 0$ and has a few large numbers. It depends on two parameters, the scale parameter m (median) and the shape parameter σ .



Probability density function:

$$p(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\log(x/m))^2}{2\sigma^2}}$$

mean: $me^{\frac{1}{2}\sigma^2}$

variance: $m^2 e^{\sigma^2} (1 - e^{-\sigma^2})$

(C.11)

C.3.6 Multinomial distribution

Generalization of the binomial distribution where n trials can have k outcomes, each with different probabilities p_i .

Probability function:

$$P(x_i) = n! \prod_{i=1}^k (p_i^{x_i} / x_i!)$$

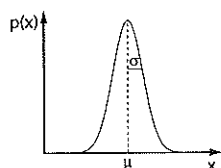
mean: np_i

variance: $np_i(1 - p_i)$

C.3.7 Normal (Gaussian) distribution

Limit of the binomial distribution for a large number of trials. Depends on two parameters, the mean μ and the standard deviation σ . The importance of the

normal distribution stems from the central limit theorem outlined below.



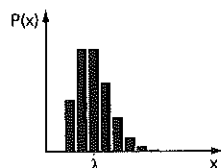
Probability density function:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

mean: μ
variance: σ^2

C.3.8 Poisson distribution

This discrete distribution is frequently used to describe the number of rare but open-ended events and to model spike trains of cortical neurons. It is closely related to the exponential distribution given above. The parameter λ is equal to the mean and the variance.



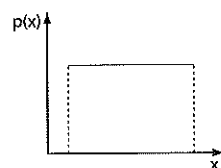
Probability function:

$$P(x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

mean: λ
variance: λ

C.3.9 Uniform distribution

Equally distributed random numbers in the interval $a \leq x \leq b$. Pseudo-random variables with this distribution are often generated by routines in many programming languages.



Probability density function:

$$p(x) = \frac{1}{b-a}$$

mean: $(a+b)/2$
variance: $(b-a)^2/12$

C.4 Cumulative probability (density) function and the Gaussian error function

The probability of having a value x for the random variable X in the range of $x_1 \leq x \leq x_2$ is given by

$$P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} p(x) dx. \quad (\text{C.12})$$

Note that we have shortened the notation by replacing the notation $P_X(x_1 \leq X \leq x_2)$ by $P(x_1 \leq X \leq x_2)$ to simplify the following expressions. In the main text we often need to calculate the probability that a normally (or Gaussian) distributed variable has values between $x_1 = 0$ and $x_2 = y$. The probability of eqn C.12 then becomes a function of y . This defines the *Gaussian error*

function

$$\frac{1}{\sqrt{2\pi}\sigma} \int_0^y e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{1}{2} \operatorname{erf}\left(\frac{y-\mu}{\sqrt{2}\sigma}\right). \quad (\text{C.13})$$

This Gaussian error function (erf) for normally distributed variables (Gaussian distribution with mean $\mu = 0$ and variance $\sigma = 1$) is commonly tabulated in books on statistics. Programming libraries also frequently include routines that return the values for specific arguments. In MATLAB this is implemented by the routine `erf`, and values for the inverse of the error function are returned by the routine `erfinv`.

Another special case of eqn C.12 is when x_1 in the equation is equal to the lowest possible value of the random variable (usually $-\infty$). The integral in eqn C.12 then corresponds to the probability that a random variable has a value smaller than a certain value, say y . This function of y is called the *cumulative density function* (cdf),²

$$P^{\text{cum}}(x < y) = \int_{-\infty}^y p(x) dx, \quad (\text{C.14})$$

which we will utilize further below.

²Note that this is a probability function, not a density function.

C.5 Functions of random variables and the central limit theorem

A function of a random variable X ,

$$Y = f(X), \quad (\text{C.15})$$

is also a random variable, Y , and we often need to know what the pdf of this new random variable is. Calculating with functions of random variables is a bit different to regular functions and some care has to be given in such situations. Let us illustrate how to do this with an example. Say we have an equally distributed random variable X as commonly approximated with pseudo-random number generators on a computer. The probability density function of this variable is given by

$$p_X(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{C.16})$$

We are seeking the probability density function $p_Y(y)$ of the random variable

$$Y = e^{-X^2}. \quad (\text{C.17})$$

The random number Y is **not** Gaussian distributed as we might think naively. To calculate the probability density function we can employ the cumulative density function eqn C.14 by noting that

$$P(Y \leq y) = P(e^{-X^2} \leq y) = P(X \geq \sqrt{-\ln y}). \quad (\text{C.18})$$

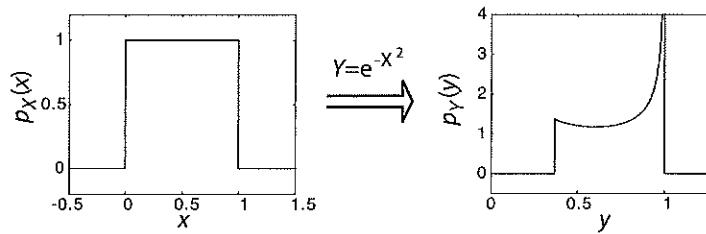
Thus, the cumulative probability function of Y can be calculated from the cumulative probability function of X ,

$$P(X \geq \sqrt{-\ln y}) = \begin{cases} \int_{\sqrt{-\ln y}}^1 p_X(x) dy = 1 - \sqrt{-\ln y} & \text{for } e^{-1} \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{C.19})$$

The probability density function of Y is the the derivative of this function,

$$p_Y(y) = \begin{cases} 1 - \sqrt{-\ln y} & \text{for } e^{-1} \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{C.20})$$

The probability density functions of X and Y are shown below.



A special function of random variables, which is of particular interest it can approximate many processes in nature, is the sum of many random variables. For example, such a sum occurs if we calculate averages from measured quantities, that is,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad (\text{C.21})$$

and we are interested in the probability density function of such random variables. This function depends, of course, on the specific density function of the random variables X_i . However, there is an important observation summarized in the *central limit theorem* that we can often utilize in this book. This theorem states that the average (normalized sum) of n random variables that are drawn from any distribution with mean μ and variance σ is approximately normally distributed with mean μ and variance σ/n for a sufficiently large sample size n . The approximation is, in practice, often very good also for small sample sizes. For example, the normalized sum of only seven uniformly distributed pseudo-random numbers is often used as a pseudo-random number for a normal distribution.

C.6 Measuring the difference between distributions

An important practical consideration is how to measure the similarity of difference between two density functions, say the density function p and the density function q . Note that such a measure is a matter of definition, similar to distance measures of real numbers or functions discussed in Appendix A.2. However, a proper distance measure, d , should be zero if the items to be compared, a and b , are the same, its value should be positive otherwise, and a distance measure should be symmetrical, meaning that $d(a, b) = d(b, a)$. The following popular measure of similarity between two density functions is not symmetrical and is hence not called a distance. It is called *Kulbach-Leibler divergence* and

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$$d^{\text{KL}}(p, q) = \int p(x) \log\left(\frac{p(x)}{q(x)}\right) dx \quad (\text{C.22})$$

$$= \int p(x) \log(p(x)) dx - \int p(x) \log(q(x)) dx \quad (\text{C.23})$$

This measure is zero if $p = q$ since $\log(1) = 0$. This measure is related to the information gain or relative entropy in information theory as discussed in Appendix D.

C.6.1 Marginal, joined, and conditional distributions

When we have probability density functions of several variables, we can distinguish several cases. This is illustrated here in the case of two variables, while generalizations to more dimensions are straight forward. The most general information we can have about a process that depends on two random variables is the *joined distribution*,

$$p(x, y), \quad (\text{C.24})$$

which is a two dimensional function. The slice of this function, given the value of one variable, say y , is the *conditional distribution*,

$$p(x|y). \quad (\text{C.25})$$

If we add over all realizations of y we get the *marginal distribution*,

$$p(x) = \int p(x, y) dy. \quad (\text{C.26})$$

Two random variables x and y are called independent if

$$p(x, y) = p(x)p(y), \quad (\text{C.27})$$

which is the same as saying

$$p(x|y) = p(x). \quad (\text{C.28})$$

Note that we can decompose the joined distribution functions into the product of a conditional and a marginal distribution,

$$p(x, y) = p(x|y)p(y) = p(y|x)p(x), \quad (\text{C.29})$$

which is sometimes called the product rule. If we divide this equation by $p(x)$, we get the identity

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}, \quad (\text{C.30})$$

which is called *Bayes theorem*. This theorem is important because it tells us how to combine a *prior* knowledge, such as the expected distribution of stimuli, $P(s)$, with some evidence as measured by a *likelihood*, $P(r|s)$, to get the *posterior* distribution $P(s|r)$ from which the optimal estimate (in a statistical sense) of the stimulus can be calculated with eqn 7.31. $P(r)$ is the proper normalization so that the left-hand side is again a probability.

Further reading

The book by Evans et al. (2000) is a useful collection of probability functions, but not an introduction to statistics. Wikipedia also has a good collection of probability functions.

Merran Evans, Nicholas Hastings, and Brian Peacock (2000), *Statistical distributions*, John Wiley & Sons, 3rd edition.

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Fundamentals of Computational Neuroscience

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